INSTITUTO
COLLEGE

MATH 110: General Education Math

Fall 2026

Worksheet 2

Name: _____

1. At each of two visits, a patient's blood pressure is classified as normal (N), elevated (E), or high (H). For example, NH means normal at the first visit and high at the second.

   (a) Write the sample space by listing its sample points.

   (b) Write the event A, "the classification is the same at both visits," by listing its sample points.

   (c) Write the event B, "the blood pressure is high at least once," by listing its sample points.

   (d) Find $A \cap B$ and $A \cup B$.

2. After a certain outpatient surgery, patients stay in the hospital for 1 to 5 days, so the sample space for the length of a randomly chosen patient's stay is $S = \{1, 2, 3, 4, 5\}$. The probability of a stay of x days is

$$P(x) = \frac{6 - x}{15}.$$

(a) Write out $\sum_{x \in S} P(x)$ term by term, and check that it equals 1.

(b) Let $A = \{3, 4, 5\}$ be the event "the patient stays at least 3 days." Write $P(A)$ using sigma notation, then evaluate it.

3. Two fair dice are rolled.

(a) What is the probability that the numbers the dice land on add up to at least 10?

(b) What is the probability that the numbers the dice land on differ by 2?

4. In the blood type experiment, the blood types of the residents of a city are distributed as follows.

Blood type	A+	A−	B+	B−	AB+	AB−	O+	O−
Probability	0.32	0.08	0.08	0.02	0.04	0.01	0.36	0.09

Recall that a patient can safely receive red blood cells only from a donor whose red blood cells carry no antigen that the patient's own red blood cells lack. A resident of the city is chosen at random.

(a) What is the probability that the resident's red blood cells carry exactly one of the A and B antigens?

(b) What is the probability that the resident's red blood cells carry at least one of the three antigens?

(c) What is the probability that the resident can receive red blood cells from a type O+ donor?

5. At a clinic, 30% of the patients have asthma, 20% have seasonal allergies, and 42% have at least one of the two. A patient is chosen at random. Let A be the event "the patient has asthma" and L the event "the patient has seasonal allergies."

(a) Find $P(A \cap L)$.

(b) What is the probability that the patient has exactly one of the two?

(c) Are A and L mutually exclusive? Are they independent? Explain.

6. A hospital has two backup generators, which work independently of each other. During a power outage, the first generator fails with probability 0.05, and the second fails with probability 0.08. Let F_1 be the event "the first generator fails" and F_2 the event "the second generator fails."

(a) Find $P(F_1 \cap F_2)$.

(b) What is the probability that at least one of the generators works?

(c) What is the probability that exactly one of the generators fails?

7. A hospital classifies its nurses by shift and by whether they have had a flu shot this season. The table gives the probabilities for a randomly chosen nurse.

	Has had a flu shot	Has not	Total
Day shift	0.42	0.18	0.60
Night shift	0.28	0.12	0.40
Total	0.70	0.30	1

Let D be the event "the nurse works the day shift" and F the event "the nurse has had a flu shot."

(a) Find $P(F \mid D)$ and $P(F \mid D^c)$.

(b) Are D and F independent? Explain.

(c) Find $P(D^c \mid F^c)$, and describe it in words.

(d) Find $P(D \cup F)$.

8. A fair coin is tossed 6 times. Let E_k be the event "exactly k of the tosses land on heads."

   (a) Find $P(E_k)$ for each k from 0 to 6.

   (b) Write the sum of your seven answers to part (a) using sigma notation. Explain why this sum must equal 1, and check that it does.

   (c) What is the probability of getting more heads than tails?

9. A practice quiz has 10 multiple-choice questions, and each question has 4 answer choices, exactly one of which is correct. A student guesses at random on every question. Let E_k be the event "exactly k of the student's answers are correct."

(a) Explain why this is a binomial experiment. What are n, a success, and p?

(b) Find $P(E_3)$.

(c) What is the probability that at least one of the student's answers is correct?

(d) The student passes the quiz with at least 8 correct answers. What is the probability that the student passes by guessing?

10. A box holds 20 rapid test kits, and 4 of them are defective. A nurse takes 3 kits from the box at random, one after the other, without putting any back.

(a) What is the probability that none of the 3 kits is defective?

(b) What is the probability that at least one of the 3 kits is defective?

(c) What is the probability that exactly one of the 3 kits is defective?

(d) A different clinic receives a very large shipment of kits, of which 20% are defective, so whether one kit is defective has no effect on whether another one is. A nurse there takes 3 kits at random. What is the probability that exactly one of them is defective? Compare with your answer to part (c).

11. Two unrelated residents of the city in Problem 4 are chosen at random, so their blood types are independent. We want to know how likely it is that the first resident can donate red blood cells to the second.

(a) Suppose the second resident has type O+ blood. What is the probability that the first resident can donate red blood cells to the second?

(b) What is the probability that the second resident has type O+ blood and the first resident can donate red blood cells to the second?

(c) What is the probability that the first resident can donate red blood cells to the second?